

Proposal: Identifying Brain Regions Sensitive to Lateralized Visual Stimuli in Mouse Decision-Making

Motivation & Research Significance

Understanding how distributed neural populations integrate sensory input and prior expectations to guide decisions is a fundamental challenge in neuroscience. The International Brain Laboratory (IBL) dataset[1], collected via a standardized behavioral paradigm and high-density Neuropixels recordings, provides an unprecedented opportunity to investigate these dynamics across the whole brain. While anatomical atlases (e.g., Allen CCF [2]) have mapped neurons spatially, there is no brain-wide functional atlas of stimulus and choice encoding. Specifically, it remains unclear which regions show the earliest and strongest sensitivity to lateralized visual stimuli, whether specialized decision-related pathways exist, and how sensory encoding transitions into choice signals. Answering these questions will advance our understanding of distributed versus specialized processing and has clinical relevance for conditions involving decision-making deficits (e.g., ADHD, schizophrenia).

Research Question & Hypotheses

Question: Which brain regions are most sensitive to left/right visual stimuli and choice encoding in a mouse visual decision-making task?

Hypotheses:

- H1 (Stimulus Encoding): Visual cortex (VISp, VISl), thalamus (LGN), and midbrain structures (SCs, APN, ZI) will exhibit the earliest and strongest differential responses to left vs right stimuli.
- H2 (Choice Encoding): Frontal cortex (MOs, PL, MOp) and basal ganglia (CP, SNr) will encode left vs right choices, distinct from primary sensory encoding.
- H₀ (Alternative): Stimulus- and choice-selective neurons are intermingled across regions, with no clear segregation.

Methodology & Feasibility

We will leverage the IBL brain-wide dataset. Our methods include:

- Preprocessing: Exclude incomplete trials and first 10 trials after block switches to minimize adaptation effects.
- Stimulus Encoding: Compute Modulation Index (MI) for each neuron to identify stimulus-responsive cells; identify contrastive clusters distinguishing left vs right.
- Latency Analysis: Estimate response latencies with Gaussian-filtered PSTHs to establish temporal hierarchy of selectivity.
- Choice Encoding: Fit Generalized Linear Models (GLMs) to quantify contributions of sensory and decision variables.
- Predictive Modeling: Train classifiers (logistic regression, XGBoost) or temporal models (RNNs/TCNs) to decode choices from population activity.
 - **(Nonlinear Decoders (Boosted Trees, SVMs, Deep MLPs)**
 - **Time-Resolved / Sequential Models - [3]**
 - **Temporal Convolutional Networks (TCNs) - [4]**
- Model Comparison: Benchmark linear vs nonlinear vs temporal decoders, highlighting tradeoffs between interpretability and predictive power.
- **After benchmarking with interpretable GLMs, we will explore whether nonlinear (XGBoost, SVM), temporal (RNN/TCN, GLM-HMM), and graph-based models (GNNs) provide additional explanatory power for choice decoding. If these advanced models outperform GLMs, it would suggest that nonlinear interactions, temporal integration, or network topology are essential features of how choice is represented across the brain**
- Statistical Rigor: Apply bootstrapping, confidence intervals, sensitivity analyses (vary exclusion windows), and Benjamini-Hochberg FDR correction. Report effect sizes alongside p-values.

The approach extends Neuromatch tutorials (PSTH, GLM, dimensionality reduction) and is feasible

within the ISP timeframe. Minimal goals include MI + latency analysis; ML decoding serves as a stretch goal.

Preliminary Work & Supporting Evidence

- Neuromatch tutorials and prior IBL work (Steinmetz et al., 2019; Stringer et al., 2019) demonstrate feasibility of extracting stimulus- and choice-selectivity.
- Early pilot analyses (e.g., PSTHs, MI) show differential responses across LGN and VISp.
- The dataset is well-curated and widely used, reducing barriers to reproducibility.

Expected Outcomes & Impact

- Scientific: A functional brain-wide atlas ranking regions by selectivity and latency for stimulus and choice encoding.
- Computational: Open-source, reproducible analysis pipeline building on community standards.
- Practical: Interactive dashboard/UI enabling researchers to explore selectivity and latency hierarchies across brain regions.
- Clinical relevance: Models of distributed vs specialized decision-making networks can inform understanding of neurological and psychiatric disorders.

Success Criteria

- Identification of at least 3–5 regions with significantly distinct left/right selectivity (FDR-corrected).
- Demonstrated latency hierarchy (>50 ms separation between early sensory and later decision-related regions).
- Decoder achieving $\geq 70\%$ accuracy on held-out data.
- Public release of pipeline and dashboard.

Proposed Work & Timeline (6 months)

- Month 1: Preprocessing + exploratory analyses (trial filtering, basic PSTHs).
- Month 2: Stimulus selectivity (MI, contrastive clustering).
- Month 3: Latency analyses across regions; draft preliminary figure.
- Month 4: Choice encoding with GLMs; test alternative models.
- Month 5: Predictive modeling with ML classifiers and RNNs (stretch goal).
- Month 6: Integration into functional atlas, dashboard development, and manuscript draft.

Summary

This project will create the first functional atlas of lateralized encoding across the mouse brain, bridging anatomy with function. By identifying when and where sensory and choice signals arise, we refine theories of distributed decision-making and provide open tools for the neuroscience community.

References

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